ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q4

Question 1

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x$ = 1280 × 400 / 3.2 = **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) LSB = $V_{range}/2^n$ = 5 V / 2^{12} = 5/4096 = 1.2207 mV.

Smallest resolvable force ΔF = LSB / S = 1.2207 mV / (10 mV/N) = **0.1221 N**.

(c) $F = e^{AT}$, $G = A^{-1}(F - I)B$. Taking T = 0.01 s.

$F = e^{-0.02} =$ **0.9802**.

G = (0.9802 − 1)/(−2) = **0.0099**.

Question 2

(a) Let $x_1 = \theta$ and $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau + b\,x_2 - mg\ell \sin x_1]$.

Equilibrium: $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Because $-1 \le \sin\theta \le 1$ we need $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) $\delta\dot{x} = A\,\delta x + B\,\delta\tau$, $A = \partial f/\partial x$, $B = \partial f/\partial \tau$ at $\theta_e = 60°$, $x_{2e} = 0$.

$A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J] = [0,\ 1;\ -122.6 \times 0.5,\ -2.5] =$ **[0, 1; −61.31, −2.5]**.

$B = [0;\ 1/J] =$ **[0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$, so the PD torque must balance gravity: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$.

If $\theta_{ss} = \theta_d = 90°$ the left side is 0 but the right side is $mg\ell \sin 90° = 2.4525 \neq 0$: contradiction. So the joint cannot rest at θ_d; the graviy torque needs a non-zero error to produce the holding torque.

Solve $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$ numerically: $\theta_{ss} \approx 1.5218$ rad = 87.19°. Error ≈ 90° − 87.19° = **2.81°** (joint sags below θ_d).

(d) With compensation, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss} = mg\ell \sin\theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; gravity cancels. Without compensation the linearised stiffness is $K_p + mg\ell\cos\theta_d$, and at $\theta_d = 90°$ $\cos\theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 4

(a) $\ddot{\theta} + a\dot{\theta} + c\theta = b(N_{ff}\,r - k_1\theta - k_2\dot{\theta}) \Rightarrow \ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$.

With a = 2, b = 1, c = 6, $k_1 = k_2 = 2$: $s^2 + 4s + 8 = 0 \Rightarrow s = -2 \pm 2j$ (stabel).

DC gain = $bN_{ff}/(c + bk_1) = N_{ff}/8 = 1 \Rightarrow$ $\mathbf{N_{ff} = 8}$.

(b) With b = 0.75 ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

$\theta_{ss} =$ **0.8**. Tracking error = (1 − 0.8)/1 × 100 = **20%**. (Less than the 25% acuator loss, because the stiffness also dropped.)

(c) Substitutting the control law and differentiating once: $s^3 + (c + bk_2)s^2 + (a + bk_1)s + bK_I$.

Desiered: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$6 + k_2 = 8 \Rightarrow$ $\mathbf{k_2 = 2}$; $2 + k_1 = 24 \Rightarrow$ $\mathbf{k_1 = 22}$; $K_I =$ **32**.

(d) For a constant r, in steady state all derivatives vanish: $\dot{\xi} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stabel so that a steady state exists.

Failure mode: integrator windup. When the actuator saturates the error persists, ξ keeps growing and causes a large overshoot afterwards. Remedy: anti-windup (clamp the integrater, conditional integration or back-calculation).